

Task no -01

Task 1 Create a bar chart or histogram to visualize the distribution of a categorical or continuous variable, such as the distribution of ages or genders in a population.

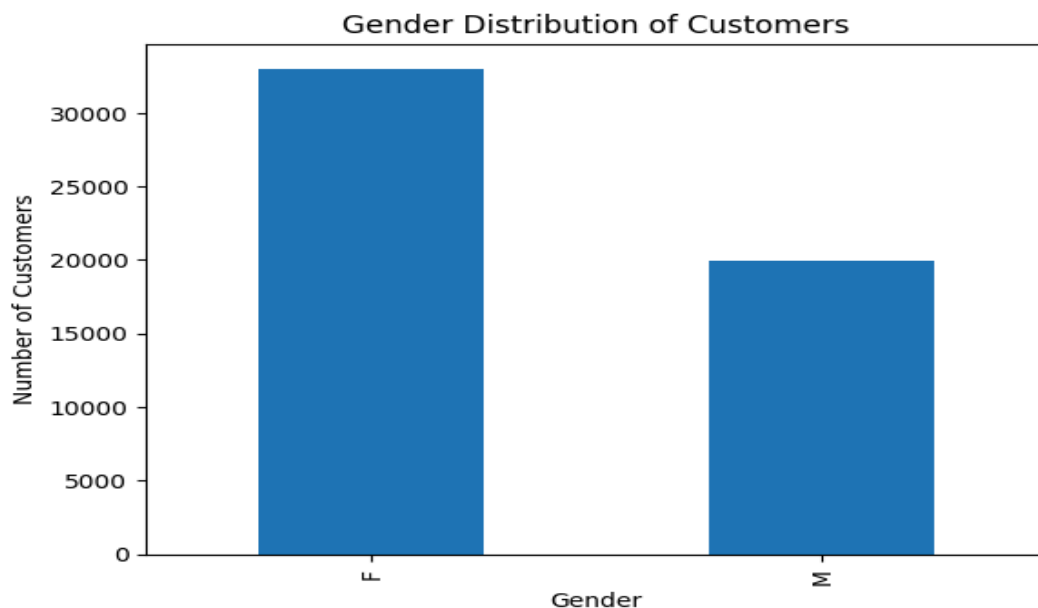
```
import numpy as np
import pandas as pd
data=pd.read_csv('/content/online shopping.csv')
data
```

Unnamed: 0	CustomerID	Gender	Location	Tenure_Months	Transaction_ID	Transaction_Date	Product_SKU	Product_Description	Product_Category	
0	0	17850	M	Chicago	12	16679	2019-01-01 00:00:00	GGOENEBJ079499	Nest Learning Thermostat 3rd Gen-USA - Stainle...	Nest-US
1	1	17850	M	Chicago	12	16680	2019-01-01 00:00:00	GGOENEBJ079499	Nest Learning Thermostat 3rd Gen-USA - Stainle...	Nest-US
2	2	17850	M	Chicago	12	16696	2019-01-01 00:00:00	GGOENEBQ078999	Nest Cam Outdoor Security Camera - USA	Nest-US

Task 1A: Bar Chart for Categorical Variable (Gender Distribution)**bold text**

```
import matplotlib.pyplot as plt
import seaborn as sns
gender_counts = data['Gender'].value_counts()

plt.figure()
gender_counts.plot(kind='bar')
plt.xlabel('Gender')
plt.ylabel('Number of Customers')
plt.title('Gender Distribution of Customers')
plt.show()
```



Task 1B: Histogram for Continuous Variable (Tenure_Months Distribution)

```
plt.figure()
plt.hist(data['Tenure_Months'], bins=10)
plt.xlabel('Tenure (Months)')
plt.ylabel('Number of Customers')
plt.title('Distribution of Customer Tenure')
plt.show()
```

